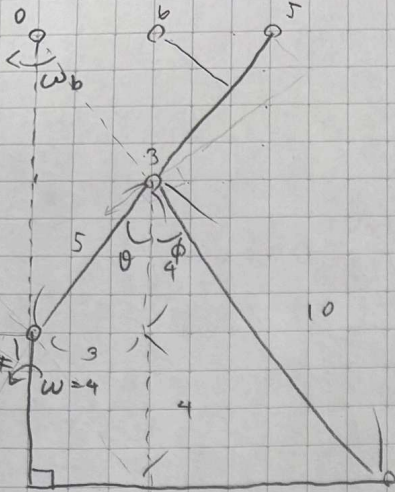
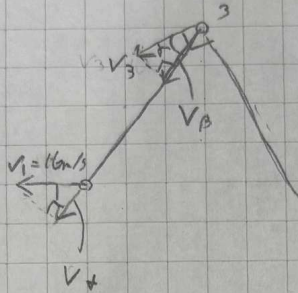


425152 回原. 修作



$$\begin{aligned} (1) \quad v_1 &= 4 \cdot 4 = 16 \text{ m/s} \\ v_1 &= 8 \cdot \omega_b = 16 \\ \omega_b &= 2 \\ v_0 &= 0 \\ v_3 &= 5 \cdot 2 = 10 \text{ m/s} \\ v_5 &= 6 \cdot 2 = 12 \text{ m/s} \\ v_6 &= 3 \cdot 2 = 6 \text{ m/s} \end{aligned}$$

(2)



図のよう v_1, v_3, v_5 とする,

$$v_1 = v_3$$

$$v_1 = v_1 \cdot \frac{3}{5}$$

$$= \frac{48}{5}$$

$$\cos \phi = \frac{8}{10} = \frac{4}{5}$$

$$\cos \theta = \frac{4}{5}$$

$$\begin{aligned} \cos(\theta + \phi) &= \frac{4}{5} \cdot \frac{4}{5} - \frac{3}{5} \cdot \frac{3}{5} \\ &= \frac{16-9}{25} = \frac{7}{25} \end{aligned}$$

$$\cos \alpha = \cos\left(\frac{\pi}{2} - \theta - \phi\right)$$

$$= \sin(\theta + \phi)$$

$$= \sqrt{1 - \frac{7^2}{25^2}}$$

$$= \sqrt{\frac{625-49}{625}} = \sqrt{\frac{576}{625}}$$

$$= \frac{24}{25}$$

$$\begin{aligned} v_3 \cdot \cos \alpha &= v_1 \\ v_3 &= \frac{16}{\frac{24}{25}} = 10 \text{ m/s} \end{aligned}$$

$$\begin{array}{r} 2 \overline{) 576} \\ 2 \overline{) 1288} \\ 2 \overline{) 144} \\ 8 \overline{) 192} \\ 9 \end{array}$$

$$\begin{aligned} 2^2 \cdot 2^2 \cdot 2^2 \cdot 3^2 \\ = 2^3 \cdot 2^3 \cdot 3^2 \end{aligned}$$